PCB DESIGN PORTFOLIO

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B.S. Electrical Engineering, Georgia Tech
Expected May 2028 · Economics minor

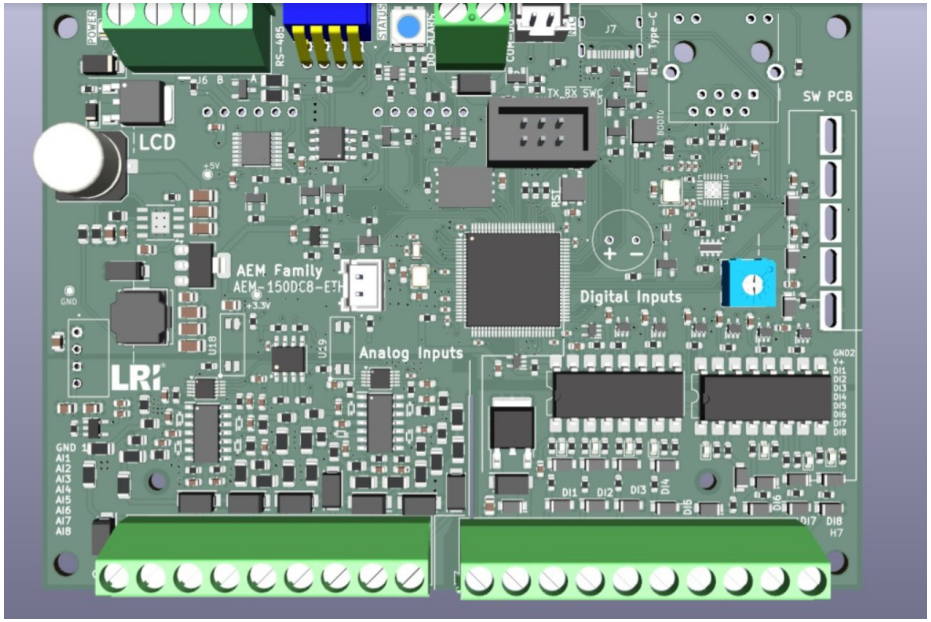
Hardware engineer focused on mixed-signal board design, power delivery and bring-up, from schematic capture through layout, fabrication and validation.

TOOLS

Altium Designer · KiCad · Cadence Allegro
SPICE / HSPICE · Oscilloscope & logic-analyzer bring-up

SOURCE FILES

github.com/akhemka-cmd/pcb-design-projects
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AEM-150DC8-ETH industrial controller, SMARK Automation (3D render)

PROJECTS

01 **SMARK AEM-150DC8-ETH** — KiCad · 2-layer
STM32 industrial I/O controller with Ethernet and isolated ADC — p. 2

02 **Automotive LiDAR — Rev 1 "Sprite"** — KiCad · 4-layer
Interface and control hardware for a scanning LiDAR — p. 4

03 **Automotive LiDAR — Rev 2** — Altium · 4-layer
Altium redesign: Teensy 4 controller, 5-ch receiver, on-board power — p. 6

04 **8S Li-ion Battery Management System** — KiCad
BQ76952 AFE, STM32 host, CAN telemetry — layout in progress — p. 8

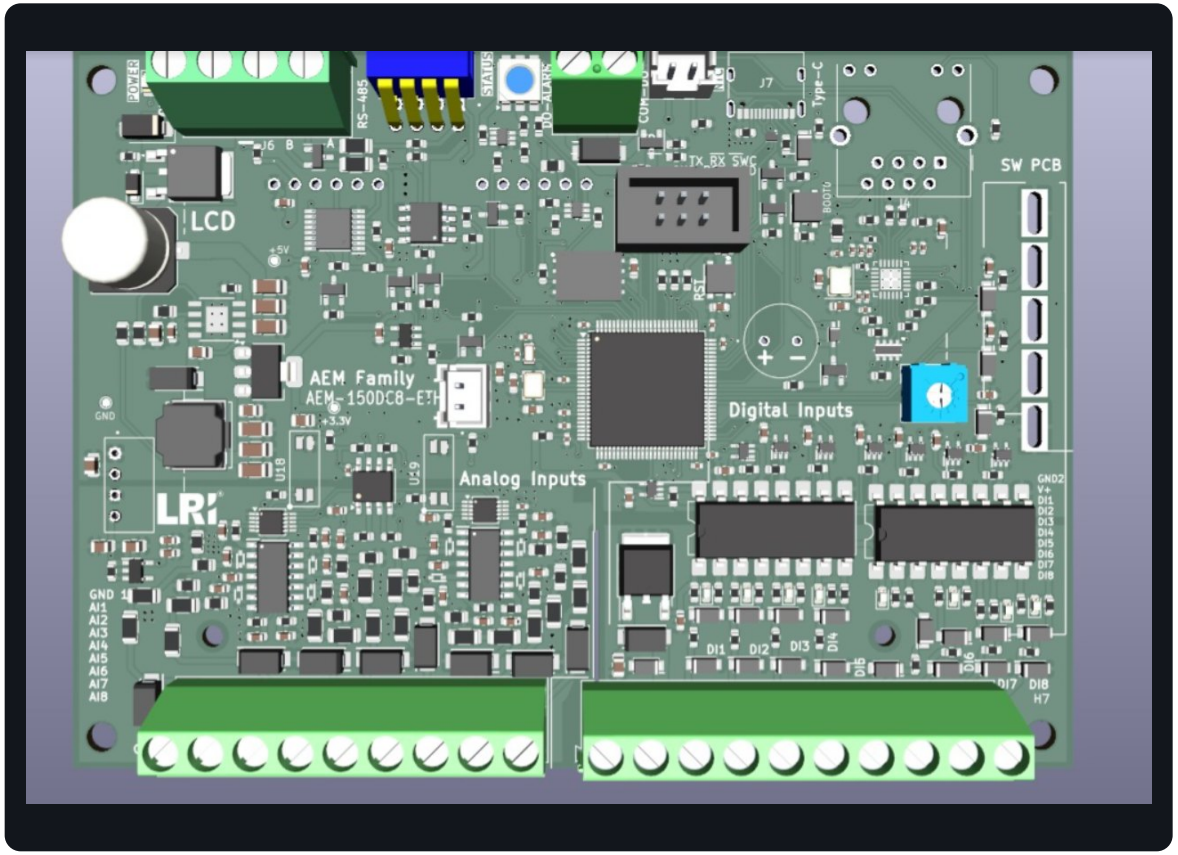
05 **12 V Power-Distribution ECU** — In development
Six-channel protected 12 V distribution with CAN-FD control — p. 10

PROJECT 01 · SMARK AUTOMATION · HARDWARE & PCB DESIGN INTERN

View source on GitHub →

AEM-150DC8-ETH Industrial I/O Controller

STM32F407-based field controller with Ethernet, RS-485 and a galvanically isolated 16-bit analog front-end



3D render of the assembled board (KiCad)

KEY SPECIFICATIONS

MCU	STM32F407VET6, Cortex-M4 (LQFP-100)
Comms	10/100 Ethernet (LAN8742A RMII PHY + RJ45), RS-485 (THVD1500), USB-C
Field I/O	8 analog inputs, 8 opto-isolated digital inputs on screw terminals
Analog	2× ADS1115 16-bit ADC, MCP6004 signal conditioning
Isolation	ISO1540 I^2C isolator + RO-0505S isolated DC-DC
Power	TPS54360 wide-input buck, linear 5 V and 3.3 V rails
Peripherals	NAND flash, EEPROM, RTC crystal, 16×4 LCD, buzzer, RGB status LED
Board	2-layer, 122 × 88 mm, ~317 placed parts, 352 vias

317
parts on the BOM

16
protected field inputs

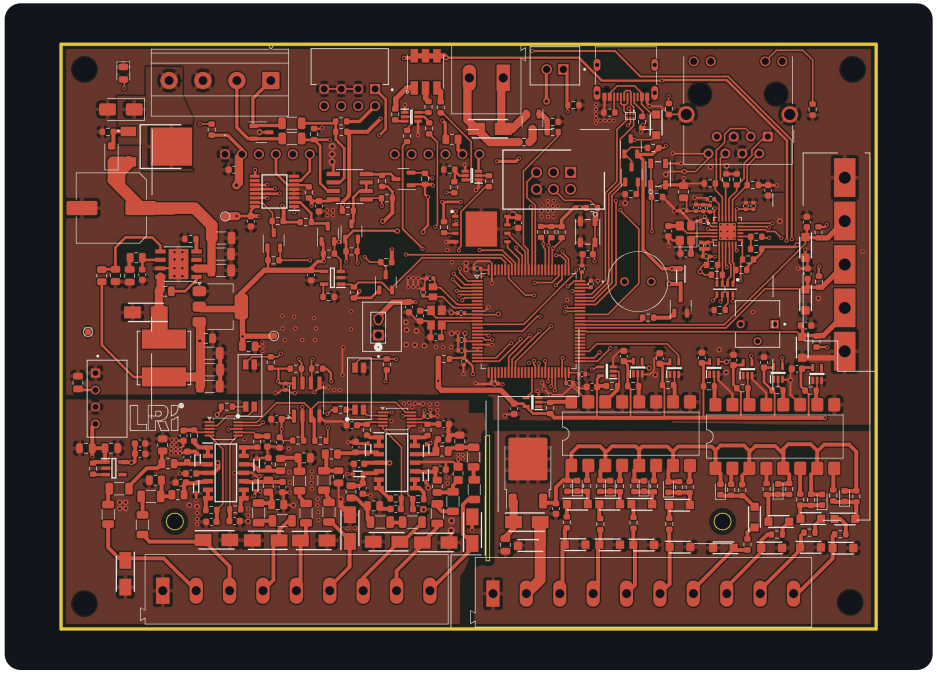
3
power domains, 2 isolated

My role: schematic capture and PCB layout in KiCad, multi-stage overvoltage protection on the field inputs, Python-based ADC validation, and a JLCPCB-ready assembly BOM.

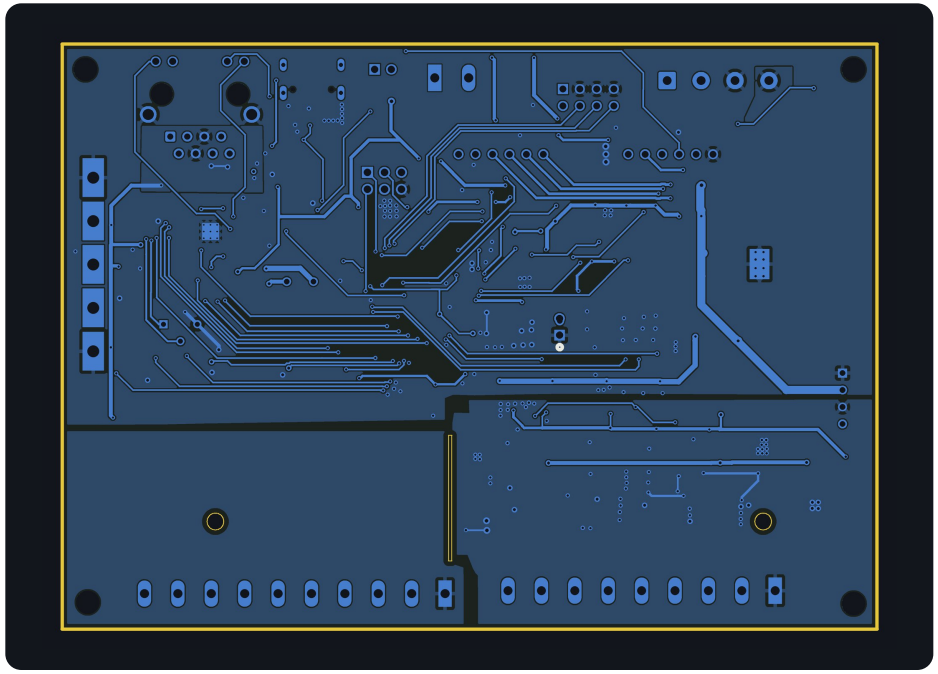
View source on GitHub →

Partitioned layout with an isolated analog island

Copper views rendered from the KiCad source



Top copper + silkscreen



Bottom copper (viewed from below)

ISOLATION & PARTITIONING

- **Galvanic isolation.** The ADC front-end sits on its own copper island; only the ISO1540 and RO-0505S cross the split.
- **Partitioning.** Analog inputs, digital inputs and the MCU each get their own region and return path.

PROTECTION

- **Field inputs.** Multi-stage overvoltage protection on every input: TVS diodes, ESD clamps and reverse-blocking diodes.
- **Comms ports.** SM712 TVS on RS-485 and a PRTR5V0U2F ESD array on USB, placed at the connectors.

SIGNALS & MANUFACTURING

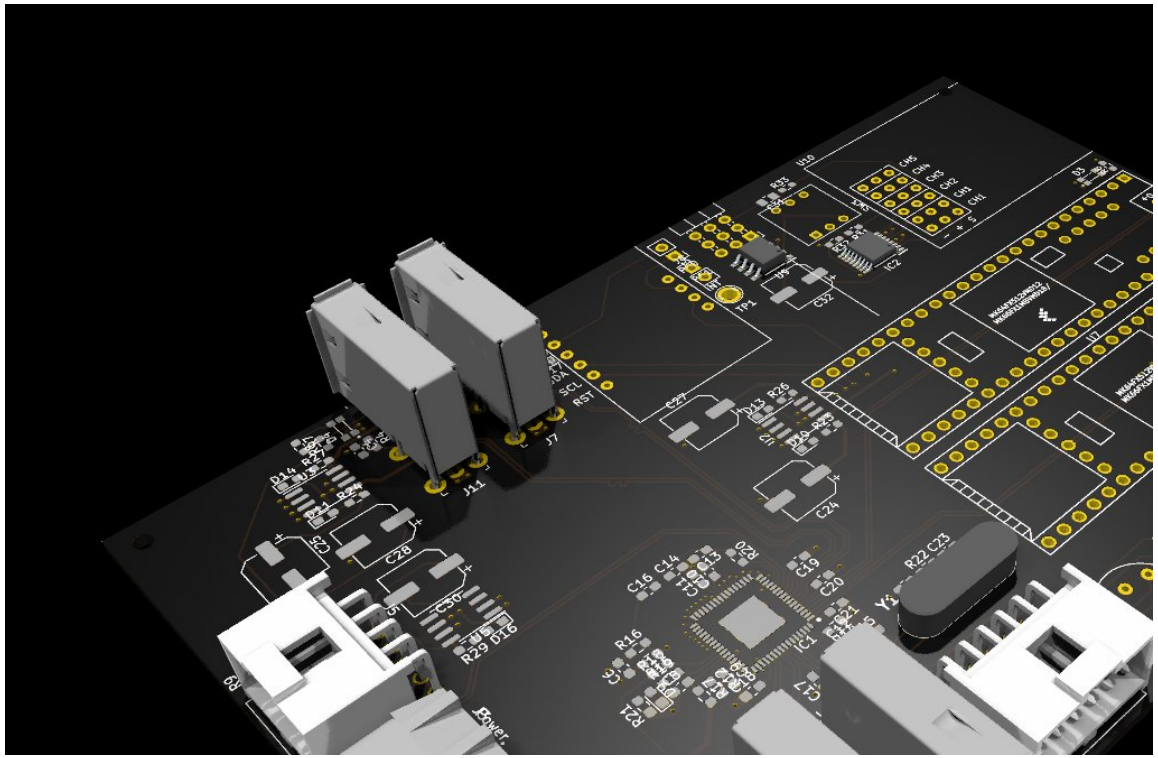
- Digital inputs pass through LTV-847 optocouplers and 74LVC1G14 Schmitt triggers for clean edges from noisy field wiring.
- LCSC part numbers and 0603 passives throughout, set up for JLCPCB turnkey assembly.

PROJECT 02 · GEORGIA TECH VIP · AUTOMOTIVE LIDAR

View source on GitHub →

Rev 1 "Project Sprite" Interface Board

Hierarchical KiCad design covering sensing, motion control, power and telemetry for a scanning LiDAR



3D render of the interface board (KiCad)

KEY SPECIFICATIONS

Controllers	Teensy 3.5 + Teensy 3.6 for the L and M control subsystems
USB	USB2517 hub (24 MHz crystal) with MIC2026 per-port power switching
Sensing	BNO055 9-axis IMU, LiDAR receiver interface
Control	74LVC157 mux for manual-override path, LM358 signal stage
Power	Board power distribution, AP2112 3.3 V LDO
Board	4-layer (2 inner power planes), 131 × 90 mm, 146 vias

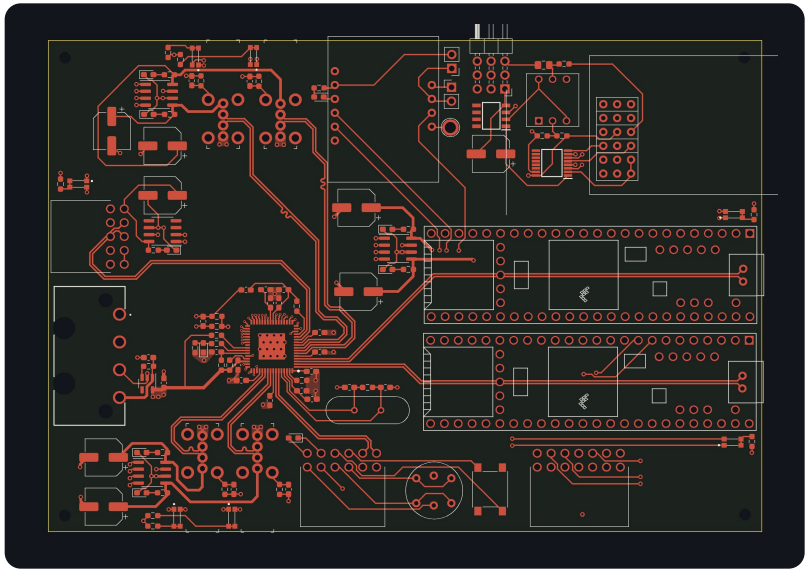
ARCHITECTURE

Four top-level sheets (Interface Board, Motion Platform, Power Supply, Scanning Mechanism) with the Interface Board split into IMU, L/M control, manual override, receiver and USB hub sub-sheets. Custom symbol, footprint and 3D libraries were built for the Teensy, hub and connectors.

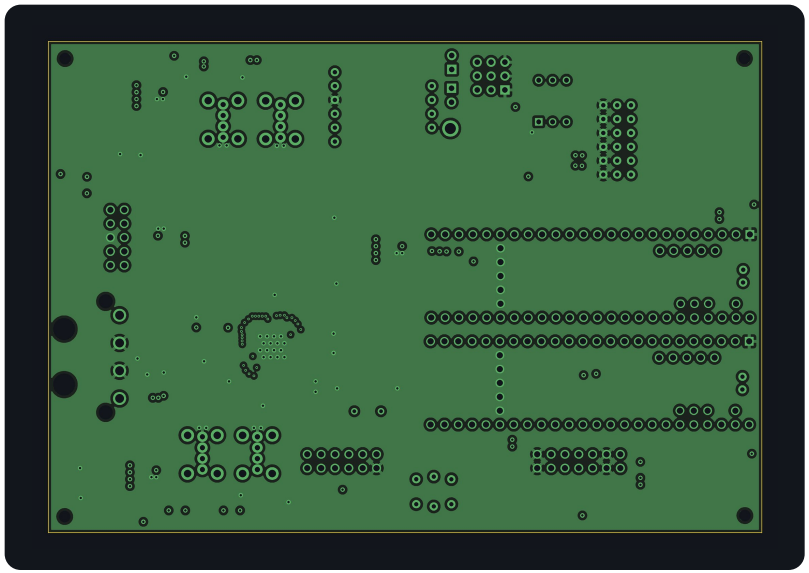
View source on GitHub →

Four-layer stackup with dedicated power planes

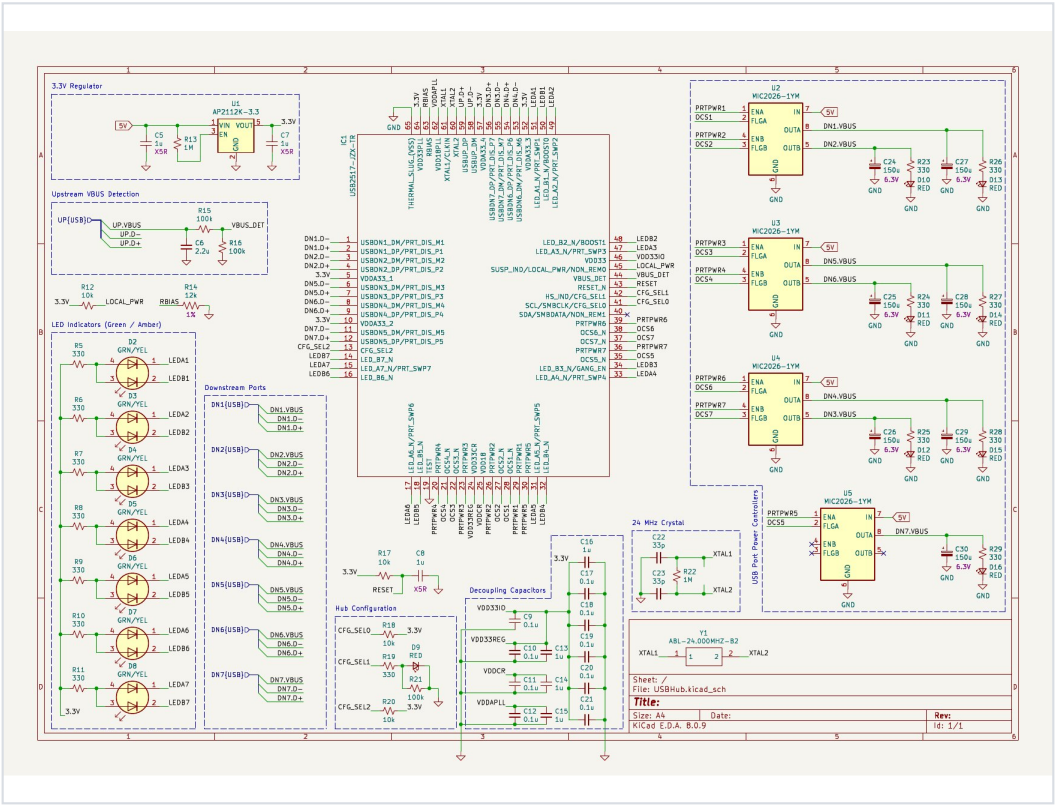
Copper views rendered from the KiCad source; schematic from the repo



Top copper + silkscreen



Inner layer 1 (power plane)



USB hub sheet: USB2517 with MIC2026 switches, upstream VBUS detection, status LEDs and decoupling

DESIGN NOTES

- Inner layers carry power planes so the top and bottom layers stay free for signal routing between the Teensy headers and connectors.
- Each downstream USB port has its own current-limited switch, so one faulty peripheral cannot take down the hub.

PROJECT 03 · GEORGIA TECH VIP · SPRING 2025 · ELECTRICAL TEAM LEAD

View source on GitHub →

Rev 2 Altium Redesign

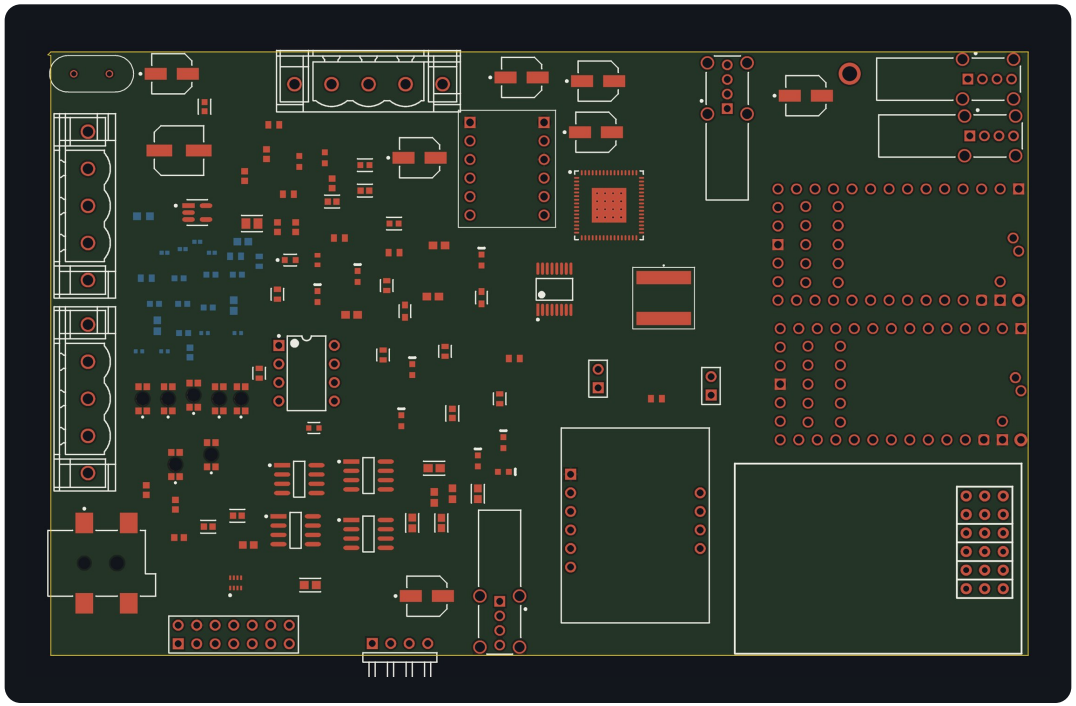
Rebuilt the LiDAR controller in Altium with a faster MCU, on-board power regulation and a 5-channel receiver

~35%
fewer inter-board connector pins

600 MHz
MCU clock, up from 120 MHz

ROS2
integration enabled

4-layer
Altium Designer



Component placement with inner-plane pour (from the Altium source)

WHAT CHANGED FROM REV 1

	Rev 1	Rev 2
EDA tool	KiCad	Altium Designer
Controller	Teensy 3.5 + 3.6	Teensy 4.0 (Cortex-M7)
Receiver	Receiver interface	5-channel receiver front-end
Power	3.3 V LDO	Dedicated 5 V regulator + 3.3 V LDO
Input	—	Barrel-jack DC input
Libraries	Custom KiCad libs	Integrated library (.IntLib)

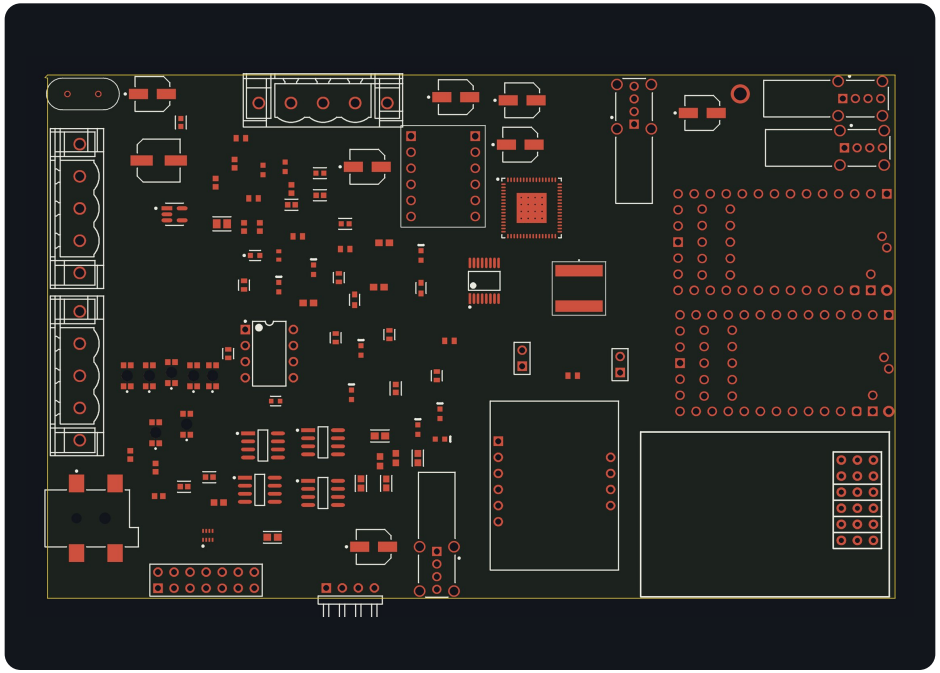
DESIGN DECISIONS

- **MCU migration.** Moved from the Teensy 3.x boards to a Teensy 4 to get the processing headroom needed for ROS2 integration.
- **Power on board.** Integrated buck-boost and LDO power management on the PCB, with power stages checked in SPICE before bring-up.
- **Motion control.** Motor control with BNO055 IMU feedback for the scanning mechanism.

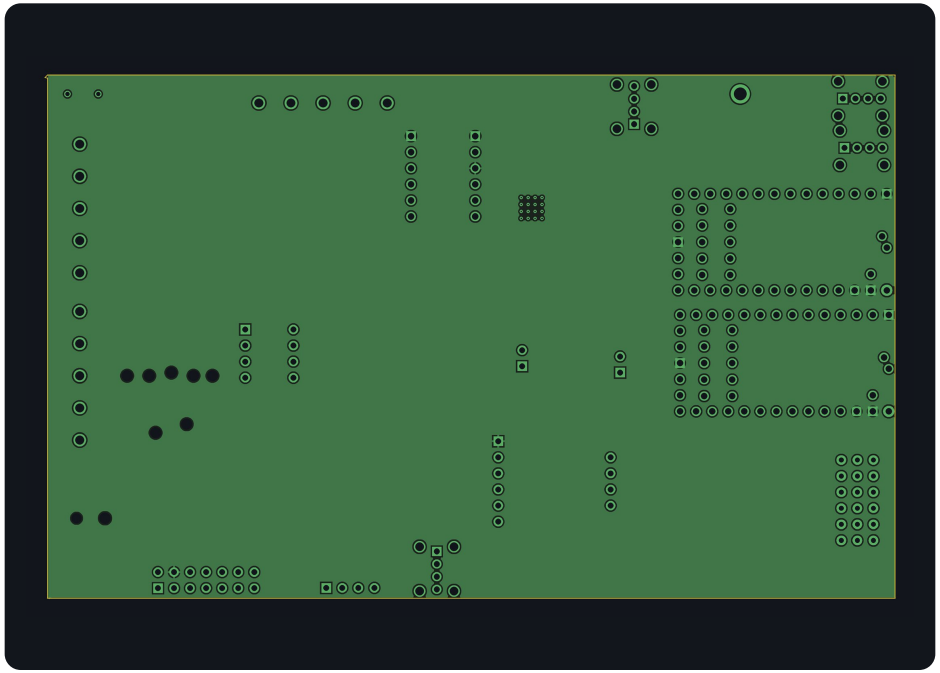
View source on GitHub →

Rev 2 layer views

Rendered from the Altium source in the repo



Top side: placement and pads



Inner layer 1: plane pour

DESIGN NOTES

- Full hierarchical Altium project: schematic and PCB libraries plus an integrated library, so the design opens with no missing parts.
- Retains the USB2517 hub with MIC2026 per-port power switching from Rev 1, adds level shifting between the Teensy and 5 V peripherals.

BRING-UP

- 4-layer board bring-up and validation as electrical team lead; 12 V-to-3.3 V buck-boost + LDO stage measured at 6 mV p-p ripple (0.5 A, 20 MHz BW).

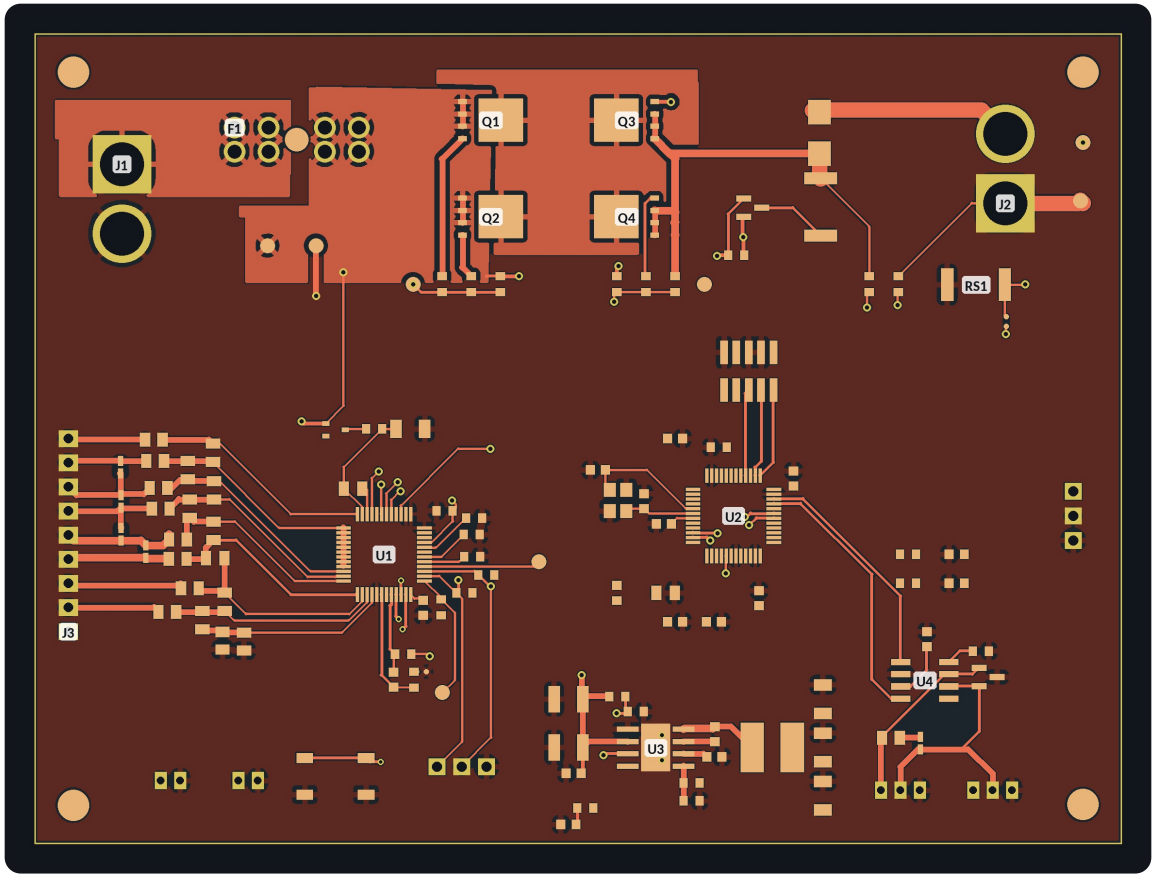
PROJECT 04 · PERSONAL PROJECT · KICAD

View source on GitHub →

8S Li-ion Battery Management System

BQ76952 analog front end with an STM32G431 host, high-side FET protection and CAN telemetry

STATUS: LAYOUT IN PROGRESS



Top copper, rendered from the KiCad source (signal routing in progress)

KEY SPECIFICATIONS

AFE	TI BQ76952 (TQFP-48), cell sense, balancing, protection
Host MCU	STM32G431CBT6 over I²C, with SWD and UART
Power path	4× 80 V BSC026N08NS5, high-side back-to-back
Current sense	1 mΩ low-side shunt, Kelvin net ties
Comms	CAN (SN65HVD230) with NUP2105L ESD
Logic supply	LMR36510 buck from the pack to 3.3 V
Board	112 × 84 mm, KiCad

My role: schematic capture, component selection and PCB layout in KiCad across four sheets: AFE, power path, MCU, and power + CAN.

120
parts on the board

4–8S
cell count via jumpers

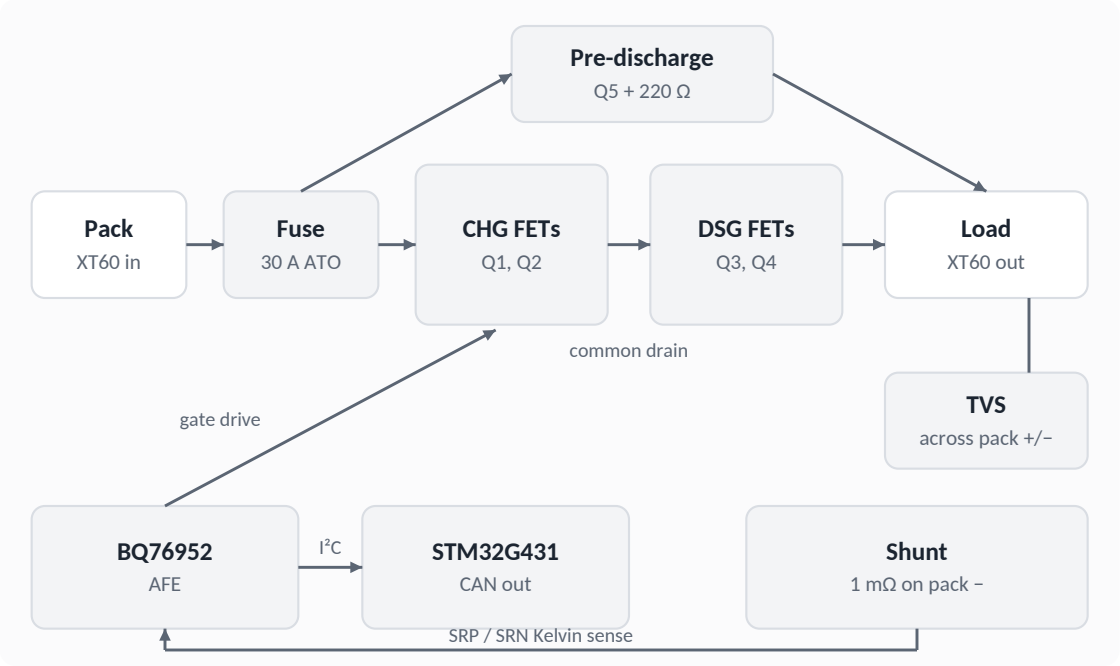
3
NTC temperature inputs

View source on GitHub →

Protection architecture

How the pack current is switched, measured and protected

POWER PATH



Block diagram drawn from the schematic. Battery negative returns through the low-side shunt.

DESIGN DECISIONS

- **High-side switching.** Back-to-back FETs on the positive rail keep pack negative continuous, so the CAN ground stays referenced when protection trips.
- **Parallel FETs.** Two 80 V parts per direction halve conduction loss and spread heat.
- **Kelvin sensing.** Net ties take the current-sense lines straight from the shunt pads.
- **Pre-discharge.** A P-FET and 220 Ω resistor charge load capacitance before DSG turns on.
- **Bring-up access.** BQStudio I²C header, SWD, UART and test points on CHG, DSG and ALERT.

NEXT STEPS

- **Routing.** Finish 3.3 V and MCU signals; replace pack-current traces with copper pours sized for the target current.
- **Bring-up and validation.** Fabricate, configure the AFE, then measure cell-voltage accuracy, protection trip timing and balancing current.

PROJECT 05 · PERSONAL PROJECT · AUTOMOTIVE POWER

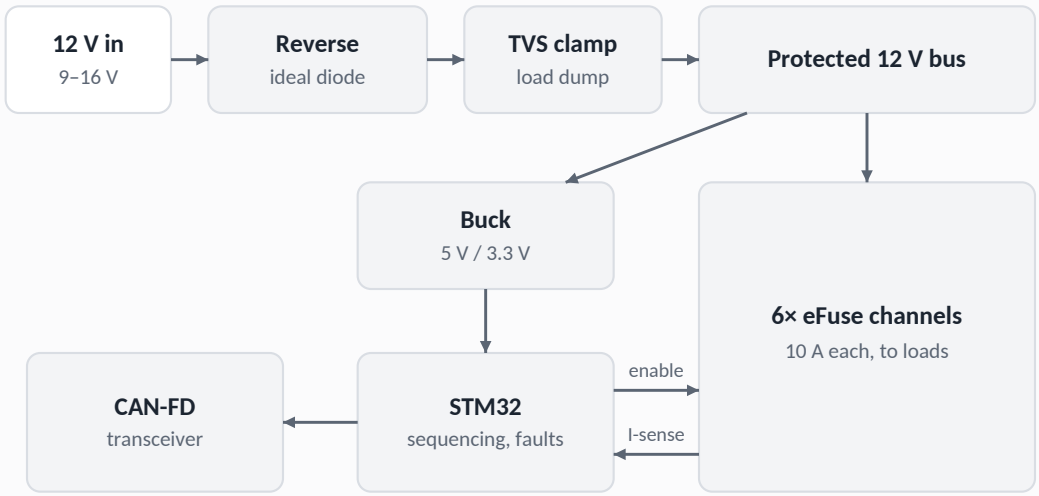
View source on GitHub →

12 V Power-Distribution ECU

Protected 12 V input, six independently switched outputs and CAN-FD control, taken from requirements to validation

STATUS: IN DEVELOPMENT — REQUIREMENTS AND ARCHITECTURE DEFINED

PLANNED ARCHITECTURE



Target architecture; part selection and schematic are the next milestones.

DESIGN REQUIREMENTS (TARGETS)

Input	9–16 V, survives reverse battery and ISO 16750-2 load dump
Outputs	6 independently switched channels, 10 A each
Sensing	Per-channel current, input voltage, board temperature
Protection	Overcurrent, short-circuit, undervoltage, thermal, with retry
Control	STM32 with CAN-FD commands and diagnostics
PCB	4-layer, copper and thermal vias sized for 10 A

VALIDATION PLAN

- **Automated bench.** Python control of supply, e-load, scope and CAN.
- **Tests.** Fault injection, current-sense accuracy, shutdown timing, full-load thermal rise.